CRITICAL CARE NURSING · LECTURE 3 · QUESTION BANK

Respiratory Disorders & Mechanical Ventilation

اضطرابات الجهاز التنفسي والتنفس الصناعي — بنك الأسئلة

235	6	13
Past exam questions	NCLEX-style	Flashcards

المصادر اللي اتجمعت منها أسئلة المحاضرة دي:

فاينال ترم أول 2023/2024 (صور الجروب) · فاينال 2021/2022 ترم تاني (صور الجروب)

فاينال 2021/2022 ترم أول (صور الجروب) · ميدترم ترم أول 2023/2024 (صور الجروب بالإجابات) · ميدترم كريتكال 2022

ميدترم كريتكال 2021 · كويز 1 كريتكال 2021 · كويز 1 كريتكال 2022 · كويز 1 كريتكال – ترم أول 2023

كويز 1 كريتكال – ترم أول 2024 · أسئلة سابقة

جنب كل سؤال مصدره. لو السؤال اتكرر في كذا امتحان مكتوب كل مرة بمصدرها، وعليه علامة "تكرر". الإجابات كلها في آخر الملف.

اللَّهُمَّ لا سَهْلَ إِلَّا ما جَعَلْتَهُ سَهْلًا، وأنتَ تَجْعَلُ الحَزْنَ إذا شِئْتَ سَهْلًا — صلِّ على النبي ﷺ

Part A · Past Exam Questions

أسئلة الدفعات السابقة

1 فاينال ترم أول 2023/2024 (صور الجروب)

Mrs. N (ARDS, intubated on AC mode). To check tube placement immediately after intubation, the nurse's responsibility is to:

a. Arrange for ECG and chest radiograph
b. Check tube placement and cuff pressure
c. Auscultate bilateral breath sounds then fix the tube
d. Fix the tube and then percuss the chest

2 فاينال ترم أول 2023/2024 (صور الجروب)

During endotracheal suctioning the nurse applies suction intermittently:

a. Only when the patient is coughing
b. While inserting the catheter
c. Only during withdrawal of the catheter
d. After insertion and while withdrawing

3 فاينال ترم أول 2023/2024 (صور الجروب)

While applying endotracheal suctioning, the recommended pressure setting is ..... mmHg.

a. 90
b. 120
c. 160
d. 190

4 During suctioning the nurse finds thick mucus difficult to remove. Which technique should be applied? فاينال ترم أول 2023/2024 (صور الجروب)

a. Administer 100% oxygen through suctioning
b. Instill normal saline before suction
c. Dry suctioning followed by saline
d. Apply chest physiotherapy and nebulization before suctioning

5 The physician ordered Diprivan (propofol) infusion for the ventilated ARDS patient in order to: فاينال ترم أول 2023/2024 (صور الجروب)

a. Liquefy the chest secretion
b. Sedate the patient
c. Expectorate the sputum
d. Aid in nebulization

6 Mr. S has ARDS: pH 7.31, PaCO2 68, HCO3 26, PaO2 60, SaO2 80%; intubated and kept prone. The earliest warning sign of ARDS is: فاينال 2021/2022 ترم تاني (صور الجروب)

a. Bilateral lung wheezing
b. Inspiratory lung crackles
c. Increased respiratory rate
d. Intercostal muscle retraction

7 This ventilated patient must be monitored for which life-threatening pulmonary complications of mechanical ventilation? (1) barotrauma (2) embolism (3) volutrauma (4) hemothorax فاينال 2021/2022 ترم تاني (صور الجروب)

a. 1,2
b. 1,3
c. 1,3 (repeated in exam)
d. 2,4

8 Prone position is recommended for this ARDS patient to: (1) reduce pressure ulcer rates (2) relieve pressure on the diaphragm (3) reduce pressure on the heart and lungs (4) distribute pleural pressure consistently فاينال 2021/2022 ترم تاني (صور الجروب)

a. 1,2,3
b. 1,2,4
c. 1,3,4
d. 2,3,4

9 Essay: Mention at least four evidence-based nursing interventions to prevent ventilator-associated pneumonia (VAP). فاينال 2021/2022 ترم تاني (صور الجروب)

.....

10 The nurse plans to have which item readily available at the bedside for a patient with respiratory distress and impending respiratory failure? فاينال 2021/2022 ترم أول (صور الجروب)

a. Crash cart
b. Intubation tray
c. Thoracentesis tray
d. Chest tube drainage system

11 Goals of mechanical ventilation do NOT include: فاينال 2021/2022 ترم أول (صور الجروب)

a. Improve lung mechanics
b. Preventing lung infection
c. Relieve excessive work of breathing
d. Improving oxygenation and removing CO2

12 One of the recommended strategies to reduce airflow resistance during mechanical ventilation is: فاينال 2021/2022 ترم أول (صور الجروب)

a. Reduce length of the ET tube
b. Use the smallest ET tube possible
c. Add water to the ventilator circuit
d. Remove the secretions in the ET tube

13 Which condition causes respiratory failure primarily by hypoventilation? فاينال 2021/2022 ترم أول (صور الجروب)

a. Pneumonia
b. Pulmonary fibrosis
c. Myasthenia gravis
d. Pulmonary embolism

14 The major cause of indirect lung injury in ARDS is: تكرر في 3 مصادر فاينال 2021/2022 ترم أول (صور الجروب)

a. Sepsis
b. Pneumonia
c. Fat embolism
d. Oxygen toxicity

15 A nurse is caring for a male patient with ARDS. Which would the nurse expect to note? فاينال 2021/2022 ترم أول (صور الجروب)

a. Pallor
b. Low arterial PaO2
c. Elevated arterial PaO2
d. Decreased respiratory rate

16 Which is a common clinical manifestation of pulmonary embolism in a male adult patient? فاينال 2021/2022 ترم أول (صور الجروب)

a. Dyspnea
b. Bradypnea
c. Bradycardia
d. Decreased respirations

17 Mr. A, 60 years, intubated for ARDS on assisted mode (rate 12, VT 400, FiO2 30%, PEEP 10). The nurse would expect which early sign of developing ARDS? ميدترم ترم أول 2023/2024 (صور الجروب بالإجابات)

a. Bilateral wheezing
b. Rapid breathing
c. Intercostal retractions
d. Paradoxical movement

18 Mr. A's ABG: pH 7.12, PaO2 70, HCO3 19, PaCO2 50, FiO2 60%. The P/F ratio is ..... and he is in ..... stage of ARDS. ميدترم ترم أول 2023/2024 (صور الجروب بالإجابات)

a. 80 & severe
b. 70 & severe
c. 220 & mild
d. 116 & moderate

19 The FIRST intervention to relieve Mr. A's hypoxemia is to: ميدترم ترم أول 2023/2024 (صور الجروب بالإجابات)

a. Increase the FiO2 level
b. Obtain an order for Bi-PAP
c. Perform oropharyngeal suction
d. Consult a chest physiotherapist

20 Mr. A's chest auscultation revealed crackles. This finding may be due to: ميدترم ترم أول 2023/2024 (صور الجروب بالإجابات)

a. Bronchospasm
b. Fluid-filled alveoli
c. Airway narrowing
d. Severe desaturation

21 To assess for the development of respiratory failure, which will be most useful? ميدترم ترم أول 2023/2024 (صور الجروب بالإجابات)

a. Bronchoscopy
b. Chest X-ray
c. Arterial blood gas analysis
d. Pulmonary function test

22 A mode of pressure-targeted ventilation that provides positive pressure to decrease workload and augment spontaneous breathing through the ET tube is: ميدترم ترم أول 2023/2024 (صور الجروب بالإجابات)

a. Bilevel inspiratory pressure
b. Pressure support ventilation
c. Positive end-expiratory pressure
d. Continuous positive airway pressure

23 Which mode of ventilation puts the patient at risk of respiratory alkalosis? ميدترم كريتكال 2022

a. Assist control
b. Pressure control
c. Controlled mandatory ventilation
d. Synchronous intermittent mandatory ventilation

24 What is the nurse's first action when the low tidal volume alarm on a patient's ventilator keeps sounding? ميدترم كريتكال 2022

a. Call the physician
b. Manually ventilate the patient
c. Check the ventilator's connections
d. Fill the endotracheal tube cuff with air

25 When alveolar ventilation relative to blood flow is reduced, which will occur? ميدترم كريتكال 2022

a. PaO2 and PaCO2 in the alveoli fall
b. PaO2 and PaCO2 in the alveoli rise
c. PaO2 in the alveoli falls and PaCO2 rises
d. PaO2 in the alveoli rises and PaCO2 falls

26 Which finding suggests a tracheoesophageal fistula in a patient who has a tracheostomy tube? ميدترم كريتكال 2022

a. The need for frequent suctioning
b. Aspiration of gastric contents in the suction catheter
c. The patient's skin and mucous membranes are light pink
d. Excessive secretions suctioned from the tracheostomy

27 The nurse assesses a patient receiving volume-cycled mechanical ventilation for which associated risk? ميدترم كريتكال 2022

a. Barotrauma
b. Hypercapnia
c. Hypoventilation
d. Respiratory acidosis

28 A patient's X-ray revealed bilateral white-outs indicating ARDS, which most frequently occurs as a result of: ميدترم كريتكال 2022

a. Impaired kidney function
b. Severe respiratory alkalosis
c. Cardiogenic pulmonary edema
d. Increased pulmonary capillary permeability

29 A patient on volume-controlled ventilation: peak airway pressure decreased from 25 to 10 cmH2O with no setting changes. The appropriate action is to: ميدترم كريتكال 2022

a. Auscultate chest and perform suctioning
b. Check the ventilator's circuit for system leaks
c. Increase the flow until pressure equals 52 cmH2O
d. Increase the volume until pressure equals 52 cmH2O

30 A patient was admitted to the ICU with respiratory distress and impending respiratory failure. As a nurse you have to keep ..... equipment available at the bedside. ميدترم كريتكال 2021

a. A chest tube
b. Intubation
c. Thoracocentesis
d. Tracheostomy

31 In a respiratory failure patient with ventilation-perfusion mismatch, which indicates that ventilation exceeds blood flow? ميدترم كريتكال 2021

a. Normal V/Q
b. High V/Q
c. Low V/Q
d. Silent V/Q

32 The major cause of indirect lung injury in ARDS is: تكرر في 3 مصادر ميدترم كريتكال 2021

a. Chest trauma
b. Severe sepsis
c. Oxygen toxicity
d. Pulmonary edema

33 The earliest sign of ARDS in a patient with multiple trauma is: ميدترم كريتكال 2021

a. Inspiratory crackles
b. Intercostal retractions
c. Increased respiratory rate
d. Bilateral wheezing

34 The standard volume- and pressure-control ventilation modes typically use an I/E ratio of: ميدترم كريتكال 2021

a. 1:1
b. 1:2
c. 1:3
d. 2:1

35 ABG for a patient with ARDS revealed pH 7.12, PaO2 70 mmHg, HCO3 19, PaCO2 50 and FiO2 60%. The PaO2/FiO2 ratio and disease stage will be: ميدترم كريتكال 2021

a. 80, severe ARDS
b. 70, severe ARDS
c. 116, moderate ARDS
d. 220, mild ARDS

36 Which condition causes respiratory failure secondary to hypoventilation? میدترم کریتکال 2021

a. Pneumonia
b. Myasthenia gravis
c. Pulmonary fibrosis
d. Pulmonary embolism

37 To reduce airflow resistance in mechanically ventilated patients, it is necessary to: میدترم کریتکال 2021

a. Lengthen the ET tube
b. Use a small ET tube size
c. Remove the ET tube secretions
d. Add water to the ventilator circuit

38 The best explanation of refractory hypoxemia is that it: میدترم کریتکال 2021

a. Accompanies restrictive airway disease
b. Persists even with 100% oxygen administration
c. Continues after weaning from oxygen therapy
d. Results from lung damage due to mechanical ventilation

39 Which of the following statements describes tidal volume? کویز 1 کریتکال 2021

a. The amount of air delivered with each breath
b. The total volume moving in and out of the lung in one minute
c. The percentage of air delivered with each breath
d. The frequency of breaths delivered by the ventilator per minute

40 Which statement indicates an understanding of pressure support ventilation (PSV) mode? کویز 1 کریتکال 2021

a. Keeps the alveoli open and prevents atelectasis
b. Allows preset pressure delivered during spontaneous ventilation
c. Achieves minimal minute ventilation
d. Delivers a preset ventilatory rate and tidal volume

41 Which should be the FIRST nursing action when assessing a patient with chronic lung disease with dyspnea? کویز 1 کریتکال 2021

a. Assess vital signs and oxygen saturation
b. Attempt to calm and reassure the patient
c. Monitor the patient every 10–15 minutes
d. Notify the health care provider immediately

42 A mode of mechanical ventilation that provides a mandatory set tidal volume and rate with each breath is: کویز 1 کریتکال 2021

a. Assist-control ventilation
b. Positive end-expiratory pressure
c. Intermittent mandatory ventilation
d. Continuous positive airway pressure

43 **Which nursing intervention should be included in the plan of care for a patient with respiratory distress?** کوئز 1 کریٹکال 2021

a. Monitor oxygen saturation every 6 hours
b. Encourage coughing and deep breathing every two days
c. Elevate the head of bed
d. Assess respiratory rate and breath sounds every 4 hours

44 **Minute ventilation is equal to:** کوئز 1 کریٹکال 2021

a. FiO2 × PEEP
b. FiO2 / PEEP
c. Tidal volume × respiratory rate
d. Tidal volume / respiratory rate

45 **Which of the following is a complication of keeping high PEEP?** کوئز 1 کریٹکال 2021

a. Hypertension
b. Hypotension
c. Hyperthermia
d. Hypothermia

46 **..... is NOT included in the care of a mechanically ventilated patient to prevent ventilator-associated pneumonia.** کوئز 1 کریٹکال 2022

a. Peptic ulcer disease prophylaxis
b. Daily oral care with chlorhexidine
c. Administering continuous sedation
d. Head of bed elevation to 30–45 degrees

47 **A patient whose pH 7.12, PaO2 70 mmHg, PaCO2 50 mmHg, HCO3 19 and FiO2 60% is expected to have a P/F ratio of ..... and a ..... stage of ARDS.** کوئز 1 کریٹکال 2022

a. 70, severe
b. 220, mild
c. 80, severe
d. 116, moderate

48 **Pulmonary artery wedge pressure (PAWP) has a significant role in the diagnosis of:** کوئز 1 کریٹکال 2022

a. Cor pulmonale
b. Respiratory failure
c. Pulmonary edema
d. Hypovolemic shock

49 **CPAP may be indicated for management of a patient who is:** کوئز 1 کریٹکال 2022

a. Alert and has spontaneous breathing
b. Immediately resuscitated after cardiac arrest
c. Comatose with high oxygen requirements
d. Requiring frequent suctioning for secretions

50 **The main cause of a high-pressure alarm in a mechanically ventilated patient is:** کوئز 1 کریٹکال 2022

a. Patient disconnection
b. Kinking of connecting tubes
c. Underinflated or ruptured cuff
d. Presence of a connection cut or hole

51 The mechanical ventilation mode which allows a patient to breathe spontaneously at his own rate and VT in between the mandatory breaths is: كويز 1 كريتكال 2022

a. Controlled ventilation
b. Assist/control ventilation
c. Positive end-expiratory pressure
d. Synchronized intermittent mandatory ventilation

52 Application of a lung-protective strategy for ARDS mechanically ventilated patients is based on: (1) high VT (2) low plateau pressure (3) FiO2 < 60% (4) PEEP كويز 1 كريتكال 2022

a. 1,2,3
b. 1,3,4
c. 2,3,4
d. 1,2,4

53 The nurse interprets a pulmonary capillary wedge pressure reading of 12 mmHg in a patient with ARDS as: كويز 1 كريتكال 2022

a. High and expected
b. Low and unexpected
c. Normal and expected
d. Uncertain and unexpected

54 The primary diagnostic criterion of ARDS is: كويز 1 كريتكال 2022

a. Inspiratory wheezing
b. Bilateral infiltrates
c. Intercostal retractions
d. Decreased respiratory rate

55 Which would the nurse expect to note in a patient with ARDS? كويز 1 كريتكال – ترم أول 2023

a. Elevated arterial SaO2
b. Low arterial PaO2
c. Elevated arterial PaO2
d. Decreased respiratory rate

56 When pH 7.12, PaO2 70 mmHg, HCO3 19, PaCO2 50 and FiO2 60%, the expected P/F ratio is ..... and the patient has ..... stage of ARDS. كويز 1 كريتكال – ترم أول 2023

a. 80 & severe
b. 70 & severe
c. 220 & mild
d. 116 & moderate

57 The major cause of indirect lung injury in ARDS is: تكرر في 3 مصادر كويز 1 كريتكال – ترم أول 2023

a. Sepsis
b. Pneumonia
c. Fat embolism
d. Oxygen toxicity

58 The nurse assesses a multiple-trauma patient for the EARLIEST sign of ARDS: كويز 1 كريتكال – ترم أول 2023

a. Bilateral wheezing
b. Inspiratory crackles
c. Intercostal retractions
d. Increased respiratory rate

59 Which of the following is one of the underlying causes of ARDS? كويز 1 كريتكال – ترم أول 2023

a. Prolonged renal failure
b. Acute respiratory alkalosis
c. Cardiogenic pulmonary edema
d. Increased pulmonary capillary permeability

60 Which is a sign of unsuccessful weaning in a patient on a 40-minute spontaneous breathing trial? كويز 1 كريتكال – ترم أول 2024

a. SaO2 is 94%
b. RR is 36 br/min
c. Rapid shallow breathing index is 85%
d. Tidal volume is 6 mL/kg

61 Minute ventilation is calculated by: كويز 1 كريتكال – ترم أول 2024

a. VT / RR
b. FiO2 / PEEP
c. VT × RR
d. FiO2 × PEEP

62 A ventilated patient: FiO2 40%, RR 14, VT 450 mL. ABG: pH 7.32, PaCO2 50, HCO3 27, PaO2 low. What is your most appropriate initial action? كويز 1 كريتكال – ترم أول 2024

a. Reduce the ventilation rate and PEEP
b. Increase the oxygen level on the ventilator
c. Perform measures to sedate the patient
d. Obtain an order for IV sodium bicarbonate

63 While caring for a ventilated patient, the ventilator alarms 'High Pressure'. This might be due to: كويز 1 كريتكال – ترم أول 2024

a. Very low blood pressure
b. Accumulation of bronchial secretions
c. Disconnection of the ventilator circuit
d. Occurrence of apnea

64 Assessment of a patient on CPAP for weaning reveals RR 32/min, SaO2 88% and use of accessory muscles. The suitable decision is to: كويز 1 كريتكال – ترم أول 2024

a. Increase the FiO2
b. Resume full ventilation
c. Extubate the patient
d. Continue the weaning

65 Which measure should the nurse include in the plan of care to reduce ventilator-associated pneumonia? كويز 1 كريتكال – ترم أول 2024

a. Administer high-flow oxygen
b. Administer bronchodilators
c. Keep the patient in supine position
d. Interrupt sedatives on a daily basis

66 Which one of the following is a risk of high PEEP? كويز 1 كريتكال – ترم أول 2024

a. Hypertension
b. Hyperthermia
c. Hypotension
d. Hypothermia

67 Which of the following is NOT a characteristic of ARDS? أسئلة سابقة

a. Dyspnea
b. Hypoxia
c. Hypercapnia
d. Frothy secretions

68 Mechanical ventilation is best described as: أسئلة سابقة

a. Invasive or non-invasive
b. A permanent device
c. A new technology
d. Doesn't induce lung injury

أسئلة سابقة

69 Pulmonary edema is characterized by: (1) frothy secretions (2) induced by heart failure (3) induced by bronchial asthma (4) tachypnea

a. 1,3,4 **b.** 1,2,4
c. 2,3,4

أسئلة سابقة

70 Low molecular weight heparin is used in treatment of pulmonary embolism. (True / False)

a. True **b.** False

أسئلة سابقة

71 Normal ECHO excludes pulmonary embolism. (True / False)

a. True **b.** False

أسئلة سابقة

72 COPD refers to:

a. Induced asthma **b.** Chronic bronchitis
c. Pulmonary embolism **d.** Atelectasis

أسئلة سابقة

73 ARDS is classified as advanced, mild, severe and complicated. (True / False)

a. True **b.** False

أسئلة سابقة

74 Smoking, heredity and allergens are all considered causes of COPD. (True / False)

a. True **b.** False

أسئلة سابقة

75 Intrinsic PEEP, oxygen toxicity and pneumothorax are complications of mechanical ventilation. (True / False)

a. True **b.** False

أسئلة سابقة

76 A protective lung strategy in mechanical ventilation sets VT at ..... mL/kg.

a. 1–3 **b.** 4–6
c. 7–10 **d.** 11–13

أسئلة سابقة

77 Which one is a management option for pulmonary edema?

a. Performing a thoracoscopy **b.** Administering specific antibiotics
c. Putting the patient on CPAP mode **d.** Administering anticoagulant drugs

أسئلة سابقة

78 Which are assist modes of ventilation? (1) volume control (2) pressure control (3) SIMV (4) CPAP

a. 1,3,4 **b.** 1,2,4
c. 2,3,4 **d.** 1,2,3

أسئلة سابقة

79 ..... is the treatment of choice in a hemodynamically unstable patient due to pulmonary embolism.

a. Adequate fluid intake **b.** Thrombolytic therapy
c. Anticoagulants **d.** IVC filter

80 Treatment of choice in a patient with massive pulmonary embolism is anticoagulation. (True / False) أسئلة سابقة

a. True **b.** False

81 A 65-year-old obese female, bedridden 10 days after femur fracture, presents with leg edema, palpitations, SOB and hypoxia; hemodynamically stable. The best imaging modality is: أسئلة سابقة

a. Chest X-ray **b.** Echocardiography
c. Pelvic-abdominal ultrasound **d.** CT pulmonary angiography

82 The following are lines of pulmonary embolism treatment EXCEPT: أسئلة سابقة

a. Oxygen **b.** Anticoagulant
c. Beta blocker drugs **d.** Thrombolytic therapy

83 To evaluate both oxygenation and ventilation in a patient with acute respiratory failure, the nurse uses ..... أسئلة سابقة

a. Chest x-rays **b.** Arterial blood gas (ABG) analysis
c. Pulse oximetry **d.** Hemodynamic monitoring

84 The nurse will monitor for clinical manifestations of hypercapnia when a patient in the emergency department has ..... أسئلة سابقة

a. Left-sided ventricular failure and acute pulmonary edema
b. Tachypnea and acute respiratory distress syndrome (ARDS)
c. Carbon monoxide poisoning after a house fire
d. Chest trauma and multiple rib fractures

85 A nurse is assessing with multiple trauma who is at risk for developing acute respiratory distress syndrome. The nurse assesses for which earliest sign of acute respiratory distress syndrome ? أسئلة سابقة

a. Bilateral wheezing **b.** Inspiratory crackles
c. intercostal retractions **d.** Increased respiratory rate

86 The disorders that cause respiratory failure due to increased work of breathing include which of the following ? أسئلة سابقة

a. Myasthenia gravis **b.** Cardiogenic pulmonary edema
c. Interstitial pulmonary fibrosis **d.** Amyotrophic lateral sclerosis
e. 1 and 2 only **f.** 2 and 3 only
g. 3 and 4 only **h.** 1 and 4 only

87 A nurse assesses a client's respiratory status. Which observation indicates that the client is having difficulty breathing ? أسئلة سابقة

a. Diaphragmatic breathing **b.** Use of accessory muscles
c. Pursed-lip breathing **d.** Controlled breathing

88 "To instruct a patient about the purpose of using the pursed-lip method of breathing, the nurse indicates that it: أسئلة سابقة

a. Promotes oxygen intake
b. Strengthens the diaphragm
c. Strengthens intercostal muscles
d. Promotes carbon dioxide elimination

89 Nursing intervention for the above patient will include: أسئلة سابقة

a. Keeping the patient in supine position
b. Preparing the patient for thoracentesis
c. Placing a non-occlusive dressing over the chest wound
d. Place a sterile occlusive dressing over the chest wound and tape it on three sides

90 A patient's condition has deteriorated. Changes in condition include trachea shift, absence of breath sounds on the left side, and hypotension. The nurse suspects that the patient has developed a(n) أسئلة سابقة

a. Cardiac tamponade
b. Hemothorax
c. Open pneumothorax
d. Ruptured diaphragm

91 What is the primary cause of pulmonary embolism (PE)? أسئلة سابقة

a. Arterial plaque rupture
b. Blood clots traveling to the lungs
c. Air embolism
d. Aneurysm rupture

92 The thrombus becomes an embolus when it: أسئلة سابقة

a. Stays in the same location
b. Detaches and travels in the bloodstream
c. Forms in the lungs
d. Blocks the right ventricle

93 What is the main factor in Virchow's Triad that leads to the formation of blood clots? أسئلة سابقة

a. Immobility
b. Stasis of blood flow
c. Hypercoagulability
d. Endothelial injury

94 Which of the following is a risk factor for pulmonary embolism related to pregnancy? أسئلة سابقة

a. Obesity
b. Pressure from the fetus on veins
c. Smoking
d. Hypercoagulability

95 Which of the following symptoms is most commonly associated with a pulmonary embolism? أسئلة سابقة

a. Palpitations
b. Sudden shortness of breath
c. Leg swelling
d. Nausea

96 Chest wheezing in pulmonary embolism is primarily due to: أسئلة سابقة

a. Bronchoconstriction
b. Decreased lung perfusion
c. Pulmonary hypertension
d. Oxygen toxicity

97 The pathophysiological mechanism behind low blood pressure in PE is due to: أسئلة سابقة

a. Impaired oxygenation of blood
b. Increased blood viscosity
c. Decreased cardiac output
d. Increased blood volume

98 What condition can predispose an individual to thrombophilia? أسئلة سابقة

a. Ulcerative colitis
b. Heart failure
c. Obesity
d. Increased tendency for blood clotting

99 A positive D-dimer test suggests Pulmonary embolism? أسئلة سابقة

a. true
b. false

100 Which imaging test is used to detect pulmonary embolism and is considered definitive? أسئلة سابقة

a. Chest X-ray
b. CT Pulmonary Angiography (CTPA)
c. Echocardiogram
d. Ventilation-perfusion (V/Q) scan

101 What is the first step in the diagnosis protocol for PE? أسئلة سابقة

a. Assess circulation
b. Assess airway
c. Assess breathing
d. Order imaging tests

102 What type of medication is primarily used to prevent further clot formation in PE? أسئلة سابقة

a. Anticoagulants
b. Antiplatelets
c. Thrombolytics
d. Antibiotics

103 Thrombolytic therapy is recommended for PE patients who are: أسئلة سابقة

a. Hemodynamically stable
b. Hemodynamically unstable
c. Pregnant
d. Receiving chemotherapy

104 In which of the following conditions would an Inferior Vena Cava (IVC) filter be used? أسئلة سابقة

a. Severe hemorrhage
b. Inability to undergo anticoagulation therapy
c. Right bundle branch block
d. Pulmonary hypertension

105 Which advanced treatment is used for severe PE cases requiring life support? أسئلة سابقة

a. Percutaneous thrombectomy
b. VA ECMO
c. Inferior Vena Cava (IVC) filter
d. Pulmonary embolectomy

106 Which of the following conditions is most likely to increase the risk of deep vein thrombosis (DVT)? أسئلة سابقة

a. Pregnancy
b. Chronic lung disease
c. Aortic aneurysm
d. Chest trauma

107 Which symptom is more likely to occur with a massive pulmonary embolism? أسئلة سابقة

a. Hypoxia
b. Mild wheezing
c. Increased blood pressure
d. Slow heart rate

108 What role does the right ventricle play in the heart's blood flow? أسئلة سابقة

a. Pumps oxygenated blood to the body
b. Pumps deoxygenated blood to the lungs
c. Receives oxygenated blood from the lungs
d. Receives deoxygenated blood from the body

109 Which of the following imaging tests is particularly useful when renal impairment is present? أسئلة سابقة

a. Chest X-ray
b. Ventilation-perfusion (V/Q) scan
c. Echocardiogram
d. CT Pulmonary Angiography (CTPA)

110 Which of the following is NOT a typical clinical feature of PE? أسئلة سابقة

a. Sudden shortness of breath
b. Chest pain
c. Hemoptysis
d. High blood pressure

111 What is the main reason for tachycardia in PE? أسئلة سابقة

a. Oxygen deficiency in the blood
b. Increased vascular resistance in the lungs
c. Decreased heart function
d. High blood pressure

112 How can pulmonary embolism lead to hypoxia? أسئلة سابقة

a. Blood flow to the lungs is blocked
b. The left ventricle fails to pump efficiently
c. Oxygenated blood cannot reach the body
d. There is too much blood flow to the lungs

113 What is a typical finding on an ECG in a patient with PE? أسئلة سابقة

a. Left bundle branch block
b. Sinus tachycardia
c. Normal sinus rhythm
d. ST-segment elevation

114 In severe PE, which intervention can physically remove or break up the clot? أسئلة سابقة

a. Thrombolytic therapy
b. Pulmonary embolectomy
c. Anticoagulation therapy
d. IVC filter insertion

115 Which is NOT a major risk factor for pulmonary embolism? أسئلة سابقة

a. Cancer
b. Smoking
c. Diabetes
d. Immobilization

116 Which type of heart failure can increase the risk of PE? أسئلة سابقة

a. Left-sided heart failure
b. Right-sided heart failure
c. Diastolic heart failure
d. Systolic heart failure

117 What causes chest pain in a patient with pulmonary embolism? أسئلة سابقة

a. Pulmonary infarction
b. Hypoxia
c. Increased lung ventilation
d. Decreased blood pressure

118 How does obesity increase the risk of PE? أسئلة سابقة

a. It increases hypercoagulability
b. It reduces oxygen uptake in the lungs
c. It causes pressure on the veins in the legs
d. It causes arterial plaque buildup

119 What imaging test can help rule out massive pulmonary embolism at the bedside? أسئلة سابقة

a. CT Pulmonary Angiography
b. Echocardiogram
c. Ventilation-perfusion (V/Q) scan
d. Lower limb venous duplex ultrasound

120 A normal lower limb venous duplex ultrasound can still be insufficient to exclude pulmonary embolism because: أسئلة سابقة

a. It only detects arterial clots
b. It may miss emboli that have already traveled to the lungs
c. It cannot detect deep vein thrombosis
d. It does not detect clot location Coma

121 What is the primary characteristic of Chronic Obstructive Pulmonary Disease (COPD)? أسئلة سابقة

a. Fully reversible airflow limitation
b. Partial reversible airflow limitation
c. Non-reversible airflow limitation
d. Progressive increase in lung elasticity

122 Which of the following conditions is a part of COPD? أسئلة سابقة

a. Emphysema
b. Chronic bronchitis
c. Asthma
d. Both emphysema and chronic bronchitis

123 In which condition does destruction and enlargement of alveoli occur? أسئلة سابقة

a. Chronic bronchitis
b. Emphysema
c. Acute exacerbation of COPD
d. Asthma

124 In chronic bronchitis, what leads to the narrowing of airways? أسئلة سابقة

a. Air trapping
b. Inflammation
c. Bronchoconstriction
d. Increased elasticity

125 How is emphysema characterized in terms of alveoli? أسئلة سابقة

a. Alveolar destruction with loss of elasticity
b. Increased production of mucus
c. Bronchoconstriction
d. Thickening of airway walls

126 What is the main problem with airflow in patients with emphysema? أسئلة سابقة

a. Air enters the lungs but cannot exit fully
b. Air does not enter the lungs at all
c. Air does not reach the alveoli
d. There is complete airway obstruction

127 Which of the following can result from the rupture of poly formed in emphysema? أسئلة سابقة

a. Pneumothorax
b. Asthma
c. Acute bronchitis
d. Pulmonary edema

128 Which of the following is NOT a common trigger for an acute exacerbation of COPD? أسئلة سابقة

a. Respiratory infections
b. Allergens
c. Change in diet
d. Air pollution

129 What symptom is associated with an acute exacerbation of COPD? أسئلة سابقة

a. Decreased cough
b. Increased cough and sputum production
c. Decreased dyspnea
d. Increased appetite

130 Which of the following is a typical finding in a physical examination of a COPD patient? أسئلة سابقة

a. Barrel-shaped chest
b. Decreased anteroposterior diameter
c. Normal chest expansion
d. High-pitched breath sounds

131 What diagnostic test is used to measure the amount and speed of air a patient can exhale in COPD? أسئلة سابقة

a. CT chest
b. Spirometry
c. Chest X-ray
d. Pulse oximeter

132 What is the target SpO2 range for oxygen supplementation in COPD patients? أسئلة سابقة

a. 70-80%
b. 88-92%
c. 92-95%
d. 80-90%

133 What condition might be observed on a CXR (Chest X-ray) in COPD patients? أسئلة سابقة

a. Hyperinflated lungs
b. Consolidation
c. Pleural effusion
d. Pneumonia

134 What is the role of bronchodilators in COPD management? أسئلة سابقة

a. To decrease inflammation in the lungs
b. To prevent airway constriction
c. To reduce sputum production
d. To improve blood oxygen levels

135 In which COPD condition is inhaled corticosteroids used? أسئلة سابقة

a. To reduce airway inflammation
b. To prevent bronchodilation
c. To enhance mucus production
d. To increase oxygen saturation

136 Which of the following is NOT typically part of COPD treatment? أسئلة سابقة

a. Antibiotics for infections
b. Oxygen supplementation
c. Inhaled corticosteroids
d. Insulin therapy

137 How is the management of COPD exacerbations generally approached? أسئلة سابقة

a. Immediate surgical intervention
b. Medications and oxygen therapy
c. Physical therapy only
d. Total lung removal

138 What is the primary goal of non-invasive ventilation in COPD patients during acute exacerbations? أسئلة سابقة

a. To prevent oxygenation failure
b. To improve blood sugar levels
c. To decrease sputum production
d. To increase lung volume

139 What is the risk associated with excessive oxygen supplementation in COPD patients? أسئلة سابقة

a. Respiratory arrest
b. Cardiac arrest
c. Increased sputum production
d. Decreased lung function

140 Which of the following is an important lifestyle modification to avoid acute exacerbations of COPD? أسئلة سابقة

a. Regular smoking
b. Increasing physical activity
c. Avoiding respiratory infections
d. Increasing alcohol consumption

141 What is the recommended action for COPD patients during weather changes to avoid exacerbations? أسئلة سابقة

a. Increase physical activity
b. Take prescribed medications regularly
c. Stop oxygen supplementation
d. Stay outdoors

142 Which of the following is NOT considered a prevention strategy for COPD exacerbation? أسئلة سابقة

a. Regular check-ups with the pulmonologist
b. Complete smoking cessation
c. Strict control of chronic diseases
d. Avoiding physical activity

143 Which of the following is a key symptom of acute COPD exacerbation? أسئلة سابقة

a. Decreased sputum production
b. Decreased cough frequency
c. Increased wheezing and chest tightness
d. Increased energy levels

144 What condition can bronchitis and pneumonia trigger in COPD patients? أسئلة سابقة

a. Pulmonary hypertension
b. COPD exacerbation
c. Asthma attack
d. Pneumothorax

145 Which diagnostic test is used to assess oxygen levels in the blood of a COPD patient? أسئلة سابقة

a. CXR
b. Pulse oximeter
c. CT scan
d. Spirometry

146 What effect does chronic bronchitis have on the airways? أسئلة سابقة

a. Increased airway elasticity
b. Destruction of alveoli
c. Chronic inflammation and narrowing
d. Reduced mucus production

147 Why is regular vaccination important for COPD patients? أسئلة سابقة

a. To prevent respiratory infections such as influenza and pneumonia
b. To improve lung function
c. To prevent the need for oxygen supplementation
d. To reduce bronchoconstriction

148 How does poly in emphysema contribute to complications like pneumothorax? أسئلة سابقة

a. It increases airway inflammation
b. It causes alveolar rupture due to overexpansion
c. It reduces airflow through the alveoli
d. It causes increased mucus production

149 What is the primary action of bronchodilators (Inhaled B2 agonists) in the management of COPD? أسئلة سابقة

a. To reduce airway inflammation
b. To prevent airway constriction by relaxing smooth muscles
c. To increase sputum production
d. To improve oxygen saturation in the blood

150 What is the primary function of spirometry in the diagnosis of COPD? أسئلة سابقة

a. To assess the structure of the lungs
b. To measure blood oxygen levels
c. To measure the amount and speed of air exhaled
d. To detect lung tumors Pulmonary Edema

151 What is the most common cause of cardiogenic pulmonary edema? أسئلة سابقة

a. Pneumonia
b. Acute pancreatitis
c. Left ventricular failure
d. Sepsis

152 Which of the following is NOT a direct lung injury that can lead to non-cardiogenic pulmonary edema? أسئلة سابقة

a. Pneumonia
b. Sepsis
c. Inhalational injury
d. High altitude pulmonary edema

153 Which of the following is characteristic of pulmonary edema? أسئلة سابقة

a. Sudden chest pain and loss of consciousness
b. Frothy sputum and shortness of breath
c. Pain in the lower extremities
d. Decreased lung sounds on auscultation

154 Which diagnostic test is used to assess hypoxemia and hypercapnia in pulmonary edema? أسئلة سابقة

a. Chest X-ray
b. Arterial Blood Gas (ABG)
c. Echocardiogram
d. CT Chest

155 What would you expect to see on a Chest X-ray (CXR) in a patient with pulmonary edema? أسئلة سابقة

a. Hyperinflated lungs with flat diaphragm
b. Increase in pulmonary parenchymal opacification
c. Ground glass opacification
d. Normal lung fields

156 Which of the following is most commonly associated with non-cardiogenic pulmonary edema? أسئلة سابقة

a. Myocardial infarction
b. Pneumonia
c. Hypertension
d. Congestive heart failure

157 Which is a typical symptom of pulmonary edema? أسئلة سابقة

a. Increased sputum production
b. Severe chest pain
c. Cyanosis and sudden collapse
d. Shortness of breath (dyspnea (

158 Which of the following is a characteristic finding in the physical examination of pulmonary edema? أسئلة سابقة

a. Bilateral wheezes without crepitations
b. Congested neck veins
c. Decreased respiratory rate
d. Unilateral reduced breath sounds

159 What is the role of BNP (B-type natriuretic peptide) in diagnosing pulmonary edema? أسئلة سابقة

a. Detects myocardial infarction
b. Indicates the stretch of the left atrium in case of heart failure
c. Detects bacterial infection
d. Measures oxygen levels in the blood

160 In the context of pulmonary edema, what is the purpose of using a Pulse Oximeter? أسئلة سابقة

a. To evaluate heart rhythm
b. To assess lung compliance
c. To measure oxygen saturation in the blood
d. To check for fluid buildup in the lungs

161 How is cardiogenic pulmonary edema differentiated from non-cardiogenic pulmonary edema? أسئلة سابقة

a. By using a chest X-ray
b. By using an echocardiogram
c. By measuring arterial blood gases
d. By performing a bronchoscopy

162 What is the most common cause of non-cardiogenic pulmonary edema? أسئلة سابقة

a. Pneumonia
b. Sepsis
c. Hypertensive crisis
d. Myocardial infarction

163 What medication is used to treat pulmonary edema by reducing fluid overload? أسئلة سابقة

a. Diuretics
b. Antibiotics
c. Steroids
d. Bronchodilators

164 Which treatment is used for severe pulmonary edema cases that require ventilatory support? أسئلة سابقة

a. Antihypertensive medication
b. Mechanical ventilation
c. Antibiotic therapy
d. Bronchodilator therapy

165 What does the term "frothy secretions" in pulmonary edema refer to? أسئلة سابقة

a. Blood-stained sputum
b. Mucous secretion mixed with air
c. Thick yellow sputum
d. Normal clear sputum

166 In the case of unstable hemodynamics due to pulmonary edema, which treatment is indicated? أسئلة سابقة

a. Diuretics
b. Defibrillation
c. Mechanical ventilation
d. Oxygen therapy

167 Which of the following findings on CT chest is characteristic of pulmonary edema? أسئلة سابقة

a. Hyperinflated lungs
b. Ground glass opacification
c. Consolidation
d. Pneumothorax

168 What is the key difference between cardiogenic and non-cardiogenic pulmonary edema? أسئلة سابقة

a. Cardiogenic edema responds to diuretics, whereas non-cardiogenic usually does not
b. Cardiogenic edema is caused by pneumonia, while non-cardiogenic is due to heart failure
c. Non-cardiogenic edema is treated with corticosteroids, cardiogenic is not
d. Non-cardiogenic edema typically requires mechanical ventilation, cardiogenic does not

169 Which of the following conditions is NOT a cause of non-cardiogenic pulmonary edema? أسئلة سابقة

a. Sepsis
b. High altitude pulmonary edema
c. Hypertension
d. Pneumonia

170 Which treatment is generally NOT used in the management of pulmonary edema? أسئلة سابقة

a. Oxygen therapy
b. Diuretics
c. Fluid overload management
d. Steroids

171 What is a common sign found on a CXR for patients with pulmonary edema? أسئلة سابقة

a. Increased lung parenchymal opacification
b. Hyperinflated lungs
c. Ground glass opacification
d. Pneumothorax

172 In which case is mechanical ventilation most likely required in pulmonary edema? أسئلة سابقة

a. Mild shortness of breath
b. Severe dyspnea with hypoxemia
c. Cough with little sputum production
d. No hypoxemia

173 Which of the following is NOT a recommended preventive measure for pulmonary edema? أسئلة سابقة

a. Regular check-ups with a cardiologist
b. Maintaining a healthy weight
c. Avoiding smoking
d. Consuming a high-salt diet

174 Which of the following is a key diagnostic tool for differentiating between cardiogenic and non- cardiogenic pulmonary edema? أسئلة سابقة

a. CT Chest
b. Echocardiogram
c. Arterial Blood Gas
d. Chest X-ray

175 What is the role of diuretics in the treatment of pulmonary edema? أسئلة سابقة

a. To reduce inflammation in the lungs
b. To increase oxygen saturation in the blood
c. To reduce fluid overload and relieve pressure on the heart
d. To decrease bronchospasm

176 Which of the following is a typical finding in a patient with cardiogenic pulmonary edema? أسئلة سابقة

a. Normal lung compliance
b. Increased lung interstitial fluid
c. Ground glass opacification
d. Unilateral chest pain

177 Which of the following conditions can lead to non-cardiogenic pulmonary edema? أسئلة سابقة

a. Myocardial infarction
b. Aspiration of gastric contents
c. Severe arrhythmias
d. Hypertension

178 What is the most appropriate action for a patient with pulmonary edema caused by sepsis? أسئلة سابقة

a. Diuretics to relieve fluid buildup
b. Treat the underlying infection with antibiotics
c. Administer mechanical ventilation
d. Administer corticosteroids

179 Which of the following is true about pulmonary edema treatment? أسئلة سابقة

a. Fluids are used to treat pulmonary edema
b. Oxygen therapy is indicated for all patients
c. Diuretics are contraindicated in non-cardiogenic pulmonary edema
d. Bronchodilators are the first-line treatment

180 In a patient with pulmonary edema and severe hypoxemia, what is the target SpO2 range? أسئلة سابقة

a. 80-85%
b. 88-92%
c. 95-100%
d. 75-80% Acute Respiratory Distress Syndrome (ARDS)

181 What is the primary characteristic of Acute Respiratory Distress Syndrome (ARDS)? أسئلة سابقة

a. Rapid onset of widespread inflammation in the lungs
b. Chronic inflammation of the airways
c. Sudden decrease in blood pressure
d. Enlargement of alveoli

182 Which of the following is a common systemic cause of ARDS? أسئلة سابقة

a. Pneumonia
b. Sepsis
c. Contusions (trauma)
d. Aspiration

183 Which of the following is a local (lung) cause of ARDS? أسئلة سابقة

a. Vasculitis
b. TRALI
c. Pneumonia
d. Pancreatitis

184 What is TRALI? أسئلة سابقة

a. A type of respiratory infection
b. A transfusion-related acute lung injury
c. A blood clotting disorder
d. A complication of severe asthma

185 What is the main presentation in ARDS? أسئلة سابقة

a. Hypotension and bradycardia
b. Hypoxemia and hypercapnia
c. Chest pain and tachycardia
d. Shortness of breath and dizziness

186 Which of the following is a typical finding on a chest X-ray in ARDS? أسئلة سابقة

a. Unilateral infiltrates
b. Bilateral infiltrates
c. Enlarged heart shadow
d. Pneumothorax

187 What does CT chest show in ARDS? أسئلة سابقة

a. Bilateral consolidation
b. Enlarged bronchi
c. Pneumothorax
d. Pleural effusion

188 What laboratory finding is expected in ARDS when an infection is present? أسئلة سابقة

a. Leukocytosis (increase in white blood cells)
b. Low platelets
c. Hyperkalemia
d. Decreased creatinine levels

189 What does an elevated liver enzyme profile (ALT, AST) in ARDS suggest? أسئلة سابقة

a. Liver dysfunction due to systemic causes
b. Acute renal failure
c. Severe pulmonary embolism
d. Cardiac arrhythmias

190 What does an elevated creatinine level indicate in ARDS? أسئلة سابقة

a. Kidney dysfunction or acute kidney injury
b. Liver failure
c. Respiratory alkalosis
d. Chronic dehydration

191 In ARDS, which autoantibody test is likely to be ordered? أسئلة سابقة

a. ANA (Antinuclear antibody)
b. HIV antibody test
c. D-dimer
d. Serum electrolytes

192 Which of the following is NOT a characteristic of ARDS as per the new global definition of 2023? أسئلة سابقة

a. Hypoxemia with PaO2/FIO2 of 201-300 for mild ARDS
b. Pulmonary edema not explained by cardiogenic causes
c. Bilateral infiltrates on chest X-ray or CT
d. Hypoxia with PaO2/FIO2 > 400 for moderate ARDS

193 Which of the following ARDS classifications corresponds to severe ARDS? أسئلة سابقة

a. PaO2/FIO2 > 300
b. PaO2/FIO2 ≤ 100 with PEEP ≥ 5
c. PaO2/FIO2 between 201-300 with NIV
d. PaO2/FIO2 > 400

194 Which of the following treatments is part of the protective lung strategy in ARDS management? أسئلة سابقة

a. High tidal volumes (TV)
b. Mechanical ventilation with a low tidal volume (< 4-6 ml/kg)
c. Increased PEEP (positive end-expiratory pressure)
d. Antibiotic therapy for pneumonia

195 What is the goal of using non-invasive ventilation (NIV) in ARDS? أسئلة سابقة

a. To decrease blood pressure
b. To support lung function without intubation
c. To provide sedatives
d. To treat pneumonia

196 In ARDS management, what is the purpose of treating the underlying cause? أسئلة سابقة

a. To reduce inflammation in the lungs
b. To improve pulmonary compliance
c. To address the systemic condition that led to ARDS
d. To prevent secondary bacterial infections

197 What is the most important component of ventilator care in ARDS patients? أسئلة سابقة

a. Use of high oxygen levels
b. Low tidal volumes and proper PEEP
c. Administration of antibiotics
d. Use of steroids for inflammation

198 Which of the following can worsen ARDS in patients with pancreatitis? أسئلة سابقة

a. Mechanical ventilation
b. Surgical resection of the pancreas
c. Administration of diuretics
d. Sepsis and abdominal infection

199 What is a key difference between ARDS and cardiogenic pulmonary edema? أسئلة سابقة

a. ARDS is primarily caused by heart failure, while cardiogenic edema is caused by infections
b. ARDS is characterized by bilateral infiltrates on CXR, which is not seen in cardiogenic edema
c. ARDS patients do not require intubation, while cardiogenic edema requires mechanical ventilation
d. ARDS responds to high-volume fluid resuscitation, unlike cardiogenic pulmonary edema

200 What is a common cause of ARDS in trauma patients? أسئلة سابقة

a. Severe dehydration
b. Pneumonia
c. Pulmonary contusions
d. Asthma exacerbation

201 In ARDS, what is the role of corticosteroids? أسئلة سابقة

a. To treat the underlying infection
b. To reduce systemic inflammation
c. To reduce pulmonary edema
d. To stabilize blood pressure

202 Which of the following findings would support a diagnosis of ARDS rather than cardiogenic pulmonary edema? أسئلة سابقة

a. Presence of pleural effusion
b. Presence of bilateral infiltrates on CXR
c. Increased oxygenation despite high PEEP
d. History of recent myocardial infarction

203 What is the goal of oxygen therapy in ARDS patients? أسئلة سابقة

a. To increase blood pressure
b. To maintain oxygen saturation above 88%
c. To reduce heart rate
d. To improve kidney function

204 Which of the following is a potential complication of ARDS management? أسئلة سابقة

a. Barotrauma from mechanical ventilation
b. Elevated platelet counts
c. Decreased white blood cell count
d. Liver dysfunction

أسئلة سابقة

205 What is the mortality rate of ARDS in most experience centers worldwide?

a. 10-20%
b. 30-40%
c. 40-50%
d. 60-70%

أسئلة سابقة

206 Which of the following investigations can identify arrhythmias that could lead to embolic stroke?

a. CT scan
b. Duplex Ultrasound
c. Electrocardiogram (ECG)
d. MRI Mechanical Ventilation:

أسئلة سابقة

207 What is the primary purpose of mechanical ventilation (MV)?

a. To directly treat illnesses
b. To help a patient breathe when they cannot breathe on their own
c. To decrease blood pressure
d. To eliminate toxins from the blood

أسئلة سابقة

208 Which of the following is a non-invasive form of mechanical ventilation?

a. Intubation
b. High Flow Nasal Cannula (HFNC)
c. Tracheostomy
d. Invasive Positive Pressure Ventilation (IPPV)

أسئلة سابقة

209 Which condition is an indication for mechanical ventilation?

a. Mild Asthma exacerbation
b. Insufficient oxygenation (hypoxemia)
c. Controlled diabetes
d. Minor allergic reactions

أسئلة سابقة

210 What is the primary benefit of mechanical ventilation in cases of insufficient ventilation?

a. It helps in removing excess oxygen
b. It aids in the removal of carbon dioxide (CO_2)
c. It helps reduce airway inflammation
d. It increases the amount of oxygen absorbed by the lungs

أسئلة سابقة

211 In which of the following situations would mechanical ventilation be needed for airway protection?

a. Mild respiratory infection
b. Patient at risk of aspiration
c. Stable asthmatic patient
d. After a minor trauma

أسئلة سابقة

212 Barotrauma is a complication of mechanical ventilation. What causes it?

a. Injury from excessive tidal volume and airway pressure
b. High oxygen levels causing tissue damage
c. Infection from the ventilator
d. Excessive CO_2 buildup in the lungs

213 **Auto-PEEP is a phenomenon in mechanical ventilation. What causes it?** أسئلة سابقة

a. Incomplete exhalation leading to positive pressure build-up
b. Excessive oxygen delivery during ventilation
c. Failure to initiate the first breath
d. Sudden reduction in ventilation rate

214 **Which of the following is not a type of mechanical ventilation?** أسئلة سابقة

a. High Flow Nasal Cannula (HFNC)
b. Helmet
c. Intubation
d. External Ventilation Mask

215 **What is the most common indication for mechanical ventilation in severe asthma cases?** أسئلة سابقة

a. Mild airway narrowing
b. Respiratory failure due to airway constriction
c. Chronic asthma management
d. Normal blood oxygen saturation levels

216 **What is the main purpose of using the Assist/Control (A/C) mode of ventilation?** أسئلة سابقة

a. To allow the patient to breathe spontaneously
b. To provide full ventilator support if the patient cannot breathe on their own
c. To help with the timing of spontaneous breaths
d. To assist the patient in weaning off mechanical support

217 **In which mode of mechanical ventilation does the ventilator provide mandatory breaths at set intervals while allowing spontaneous breathing?** أسئلة سابقة

a. Assist/Control (A/C)
b. Synchronous Intermittent Mandatory Ventilation (SIMV)
c. Pressure Support Ventilation (PSV)
d. Continuous Positive Airway Pressure (CPAP)

218 **Pressure Support Ventilation (PSV) is most appropriate for which type of patient?** أسئلة سابقة

a. A patient who is unable to breathe on their own
b. A patient who requires full ventilator support
c. A patient who can breathe independently but needs assistance with the effort
d. A patient with acute lung injury

219 **Which of the following is true regarding Continuous Positive Airway Pressure (CPAP) mode?** أسئلة سابقة

a. It provides complete control of breathing
b. It keeps the airways open without controlling the patient's breaths
c. It allows for spontaneous breathing only
d. It is used only during sleep

220 What is the main difference between Control Mode Ventilation (CMV) and other ventilation modes? أسئلة سابقة

a. The patient controls the timing of breaths
b. The ventilator provides all breaths with no spontaneous breathing allowed
c. The patient does not require any mechanical support
d. The mode alternates between high and low airway pressures

221 Which of the following is not included in the Ventilator Bundle for reducing complications like VAP? أسئلة سابقة

a. Head of bed elevation
b. Oral care with chlorhexidine
c. Daily sedation vacation
d. Increased oxygen saturation

222 What is the purpose of peptic ulcer prophylaxis in mechanically ventilated patients? أسئلة سابقة

a. To prevent respiratory infections
b. To prevent the development of stress ulcers
c. To promote lung expansion
d. To reduce inflammation in the lungs

223 Which of the following complications is most commonly associated with mechanical ventilation for extended periods? أسئلة سابقة

a. Barotrauma
b. Pneumothorax
c. Ventilator-Associated Pneumonia (VAP)
d. All of the above

224 High Flow Nasal Cannula (HFNC) provides what type of support in mechanical ventilation? أسئلة سابقة

a. High levels of oxygen delivery
b. Positive pressure ventilation
c. Complete mechanical ventilation support
d. Just oxygen delivery without pressure

225 In which situation is intubation the most appropriate intervention? أسئلة سابقة

a. A patient requiring short-term oxygen support
b. A patient with moderate asthma exacerbation
c. A patient with severe airway obstruction or respiratory failure
d. A patient needing oxygen at home

226 Which of the following is a benefit of mechanical ventilation? أسئلة سابقة

a. Decreases blood pressure
b. Increases patient activity levels
c. Facilitates carbon dioxide removal
d. Increases the production of red blood cells

227 Ventilator-Induced Lung Injury (VILI) can occur due to which of the following? أسئلة سابقة

a. High PEEP and excessive tidal volume
b. Low PEEP and low tidal volume
c. Low-frequency ventilation
d. Short-term oxygen therapy

228 The Synchronous Intermittent Mandatory Ventilation (SIMV) mode is most appropriate for patients who: أسئلة سابقة

a. Cannot breathe independently
b. Require full support for every breath
c. Are starting to breathe independently but need some ventilator assistance
d. Have complete lung collapse

229 In which of the following conditions would Mechanical Ventilation be contraindicated? أسئلة سابقة

a. Severe asthma exacerbation
b. Acute lung injury (ALI)
c. Cardiac arrest with no chance of recovery
d. Acute respiratory distress syndrome (ARDS)

230 Which ventilator mode is best suited for a patient who needs full ventilator support and cannot breathe on their own? أسئلة سابقة

a. Assist/Control (A/C)
b. Pressure Support Ventilation (PSV)
c. Synchronous Intermittent Mandatory Ventilation (SIMV)
d. Continuous Positive Airway Pressure (CPAP)

231 Which complication of mechanical ventilation results from prolonged exposure to high oxygen levels? أسئلة سابقة

a. Pneumothorax
b. Oxygen toxicity
c. Barotrauma
d. Hypercapnia

232 What is the goal of Daily Sedation Vacation in ventilated patients? أسئلة سابقة

a. To increase the level of sedation
b. To assess whether sedation can be reduced or stopped
c. To adjust the ventilator settings
d. To assess the patient's need for mechanical ventilation

233 High PEEP levels during mechanical ventilation may lead to which complication? أسئلة سابقة

a. Oxygen toxicity
b. Pneumothorax
c. Oxygen desaturation
d. Hypercapnia

234 Which of the following ventilator alarms alerts the healthcare team to a potential issue with the patient's ventilation rate? أسئلة سابقة

a. High Pressure alarm
b. High Frequency alarm
c. Low Pressure alarm
d. Apnea alarm

235 What is the recommended action if a patient is awake after ROSC? أسئلة سابقة

a. Continue CPR
b. Start targeted temperature management
c. Transfer the patient to advanced critical care
d. Administer lidocaine

Part B · NCLEX-Style Practice

أسئلة على طريقة NCLEX (إضافة)

N1 The PaO2/FiO2 ratio of 150 indicates: NCLEX

a. Mild ARDS
b. Moderate ARDS
c. Severe ARDS
d. Normal

N2 A high-pressure ventilator alarm is most often caused by: NCLEX

a. Circuit disconnection
b. Secretions, kinking or biting the tube
c. Cuff leak
d. Apnea

N3 Select ALL components of the VAP prevention bundle. SELECT ALL

a. HOB 30–45°
b. Daily sedation interruption
c. Oral care with chlorhexidine
d. Peptic ulcer & DVT prophylaxis
e. Routine daily circuit changes

N4 Lung-protective ventilation for ARDS uses a tidal volume of: NCLEX

a. 10–12 mL/kg
b. 4–6 mL/kg
c. 15 mL/kg
d. 1–2 mL/kg

N5 Suction is applied: NCLEX

a. During insertion
b. Intermittently during withdrawal, ≤10–15 s, at 80–120 mmHg
c. Continuously for 30 s
d. Only when coughing

N6 The treatment of choice for hemodynamically unstable (massive) pulmonary embolism is: NCLEX

a. Anticoagulants alone
b. Thrombolytic therapy
c. Beta-blockers
d. Fluid restriction

Part C · Quizlet Flashcards

كروت مراجعة سريعة

غطّي العمود الأبيض بورقة واختبر نفسك: شوف المصطلح وقول التعريف.

ARDS	Non-cardiogenic pulmonary edema; ↑ capillary permeability.	**ARDS earliest sign**	Increased respiratory rate.
ARDS indirect cause	Sepsis.	**ARDS diagnosis**	Bilateral infiltrates; refractory hypoxemia.

P/F ratio	Mild 200–300, moderate 100–200, severe <100.
Minute ventilation	VT × RR.
AC mode	Every breath delivers set VT; risk of resp alkalosis.
SIMV	Spontaneous breaths between mandatory ones.
PSV/CPAP	Supports spontaneous breathing; CPAP for alert patients.
High PEEP	Hypotension, barotrauma.
High-pressure alarm	Secretions, kinking, biting.
Low-pressure alarm	Disconnection, leak.
PE	Dyspnea, tachypnea; CTPA; LMWH; thrombolysis if unstable.

Answer Key

الإجابات

1. c — Auscultate bilateral breath sounds then fix the tube
إجابة مقترحة (المصدر مفيهوش إجابة رسمية). فاينال 2023/2024 من بزّه (VF، cardiac cath) غير إجابات — الإجابات مقترحة. باقي الامتحان القائمة.

2. c — Only during withdrawal of the catheter
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

3. b — 120
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

4. d — Apply chest physiotherapy and nebulization before suctioning
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

5. b — Sedate the patient
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

6. c — Increased respiratory rate

7. b — 1,3
الاختيارين b و c 1,3» في الورقة الاتنين» (barotrauma + volutrauma) الاتنين نفس المحتوى c الطلبة علّموا —.

8. d — 2,3,4

9. ✎ — HOB elevation 30–45°; daily sedation interruption & readiness-to-wean assessment; oral care with chlorhexidine; subglottic suctioning / cuff pressure 20–30 cmH2O; hand hygiene; PUD & DVT prophylaxis; avoid unnecessary circuit changes

10. b — Intubation tray

11. b — Preventing lung infection

12. d — Remove the secretions in the ET tube

13. c — Myasthenia gravis

14. a — Sepsis

15. b — Low arterial PaO2

16. a — Dyspnea
وده غلط — العرض الأشيع decreased respirations الورقة متظللة على هو ضيق النفس المفاجئ في الـ PE (dyspnea) وسرعة التنفس.

17. b — Rapid breathing

18. d — 116 & moderate

19. c — Perform oropharyngeal suction
إجابة الورقة oropharyngeal suction (أول خطوة airway). أسئلة ECG والأدوية والحروق في الميدترم ده اتشالت (بزّه القائمة).

20. b — Fluid-filled alveoli

21. c — Arterial blood gas analysis
الورقة معلّمة chest X-ray، والأدق ABG (المعيار الأساسي لتشخيص الفشل التنفسي).

22. b — Pressure support ventilation

23. a — Assist control
وأدوية ECG ميدترم 2022 فيه كمان أسئلة (propofol, beta blockers, nitroglycerin, atropine, dobutamine, thrombolytics) — اتشالت لأنها بزّه القائمة.

24. c — Check the ventilator's connections

25. c — PaO2 in the alveoli falls and PaCO2 rises

26. b — Aspiration of gastric contents in the suction catheter

27. a — Barotrauma

28. d — Increased pulmonary capillary permeability

29. b — Check the ventilator's circuit for system leaks

30. b — Intubation

31. b — High V/Q

32. b — Severe sepsis

33. c — Increased respiratory rate

34. b — 1:2

35. c — 116, moderate ARDS
PaO2/FiO2 = 70/0.6 ≈ 116.7 → moderate ARDS (100–200).

36. b — Myasthenia gravis

37. c — Remove the ET tube secretions

38. b — Persists even with 100% oxygen administration

39. a — The amount of air delivered with each breath

40. b — Allows preset pressure delivered during spontaneous ventilation

41. a — Assess vital signs and oxygen saturation
أسئلة الأدوية (dopamine, nitroglycerin, atropine) في كويز 1 كريتكال 2021. و2022 اتشالت لأنها بزّه القائمة.

42. a — Assist-control ventilation

43. c — Elevate the head of bed

44. c — Tidal volume × respiratory rate

45. b — Hypotension

46. c — Administering continuous sedation

47. d — 116, moderate

48. c — Pulmonary edema

49. a — Alert and has spontaneous breathing

50. b — Kinking of connecting tubes

51. d — Synchronized intermittent mandatory ventilation

52. c — 2,3,4

53. c — Normal and expected

54. b — Bilateral infiltrates

55. b — Low arterial PaO2

56. d — 116 & moderate

57. a — Sepsis

58. d — Increased respiratory rate

59. d — Increased pulmonary capillary permeability

60. b — RR is 36 br/min

61. c — VT × RR

62. b — Increase the oxygen level on the ventilator
قيمة الـ PaO2 مقصوصة في الصورة؛ الطالب اختار زيادة الأكسجين.

63. b — Accumulation of bronchial secretions

64. b — Resume full ventilation

65. d — Interrupt sedatives on a daily basis

66. c — Hypotension

67. d — Frothy secretions

68. a — Invasive or non-invasive

69. b — 1,2,4

70. a — True

71. b — False

72. b — Chronic bronchitis

73. b — False

74. a — True

75. a — True

76. b — 4–6

77. c — Putting the patient on CPAP mode

78. c — 2,3,4
في نفس الملف ACS/ACLS الإجابة المعلّمة 2,3,4 من الفورم — أسئلة اتشالت (برّه القائمة).

79. b — Thrombolytic therapy

80. b — False

81. d — CT pulmonary angiography

82. c — Beta blocker drugs

83. b — Arterial blood gas (ABG) analysis

84. d — Chest trauma and multiple rib fractures

85. d — Increased respiratory rate

86. f — 2 and 3 only

87. b — Use of accessory muscles

88. d — Promotes carbon dioxide elimination

89. d — Place a sterile occlusive dressing over the chest wound and tape it on three sides

90. b — Hemothorax

91. b — Blood clots traveling to the lungs

92. b — Detaches and travels in the bloodstream

93. b — Stasis of blood flow

94. b — Pressure from the fetus on veins

95. b — Sudden shortness of breath

96. a — Bronchoconstriction

97. c — Decreased cardiac output

98. d — Increased tendency for blood clotting

99. b — false

100. b — CT Pulmonary Angiography (CTPA)

101. b — Assess airway

102. a — Anticoagulants

103. b — Hemodynamically unstable

104. b — Inability to undergo anticoagulation therapy

105. b — VA ECMO

106. a — Pregnancy

107. a — Hypoxia

108. b — Pumps deoxygenated blood to the lungs

109. b — Ventilation-perfusion (V/Q) scan

110. d — High blood pressure

111. a — Oxygen deficiency in the blood

112. a — Blood flow to the lungs is blocked

113. b — Sinus tachycardia

114. b — Pulmonary embolectomy

115. c — Diabetes

116. b — Right-sided heart failure

117. a — Pulmonary infarction

118. c — It causes pressure on the veins in the legs

119. b — Echocardiogram

120. b — It may miss emboli that have already traveled to the lungs

121. c — Non-reversible airflow limitation

122. d — Both emphysema and chronic bronchitis

123. b — Emphysema

124. b — Inflammation

125. a — Alveolar destruction with loss of elasticity

126. a — Air enters the lungs but cannot exit fully

127. a — Pneumothorax

128. c — Change in diet

129. b — Increased cough and sputum production

130. a — Barrel-shaped chest

131. b — Spirometry

132. b — 88-92%

133. a — Hyperinflated lungs

134. b — To prevent airway constriction

135. a — To reduce airway inflammation

136. d — Insulin therapy

137. b — Medications and oxygen therapy

138. a — To prevent oxygenation failure

139. a — Respiratory arrest

140. c — Avoiding respiratory infections

141. b — Take prescribed medications regularly

142. d — Avoiding physical activity

143. c — Increased wheezing and chest tightness

144. b — COPD exacerbation

145. b — Pulse oximeter

146. c — Chronic inflammation and narrowing

147. a — To prevent respiratory infections such as influenza and pneumonia

148. b — It causes alveolar rupture due to overexpansion

149. b — To prevent airway constriction by relaxing smooth muscles

150. c — To measure the amount and speed of air exhaled

151. c — Left ventricular failure

152. b — Sepsis

153. b — Frothy sputum and shortness of breath

154. b — Arterial Blood Gas (ABG)

155. b — Increase in pulmonary parenchymal opacification

156. b — Pneumonia

157. d — Shortness of breath (dyspnea (

158. b — Congested neck veins

159. b — Indicates the stretch of the left atrium in case of heart failure

160. c — To measure oxygen saturation in the blood

161. b — By using an echocardiogram

162. b — Sepsis

163. a — Diuretics

164. b — Mechanical ventilation

165. b — Mucous secretion mixed with air

166. b — Defibrillation

167. b — Ground glass opacification

168. a — Cardiogenic edema responds to diuretics, whereas non-cardiogenic usually does not

169. c — Hypertension

170. d — Steroids

171. a — Increased lung parenchymal opacification

172. b — Severe dyspnea with hypoxemia

173. d — Consuming a high-salt diet

174. b — Echocardiogram

175. c — To reduce fluid overload and relieve pressure on the heart

176. b — Increased lung interstitial fluid

177. b — Aspiration of gastric contents

178. b — Treat the underlying infection with antibiotics

179. b — Oxygen therapy is indicated for all patients

180. b — 88-92%

181. a — Rapid onset of widespread inflammation in the lungs

182. b — Sepsis

183. c — Pneumonia

184. b — A transfusion-related acute lung injury

185. b — Hypoxemia and hypercapnia

186. b — Bilateral infiltrates

187. a — Bilateral consolidation

188. a — Leukocytosis (increase in white blood cells)

189. a — Liver dysfunction due to systemic causes

190. a — Kidney dysfunction or acute kidney injury

191. a — ANA (Antinuclear antibody)

192. d — Hypoxia with PaO2/FIO2 > 400 for moderate ARDS

193. b — PaO2/FIO2 ≤ 100 with PEEP ≥ 5

194. b — Mechanical ventilation with a low tidal volume (< 4-6 ml/kg)

195. b — To support lung function without intubation

196. c — To address the systemic condition that led to ARDS

197. b — Low tidal volumes and proper PEEP

198. d — Sepsis and abdominal infection

199. b — ARDS is characterized by bilateral infiltrates on CXR, which is not seen in cardiogenic edema

200. c — Pulmonary contusions

201. b — To reduce systemic inflammation

202. b — Presence of bilateral infiltrates on CXR

203. b — To maintain oxygen saturation above 88%

204. a — Barotrauma from mechanical ventilation

205. c — 40-50%

206. c — Electrocardiogram (ECG)

207. b — To help a patient breathe when they cannot breathe on their own

208. b — High Flow Nasal Cannula (HFNC)

209. b — Insufficient oxygenation (hypoxemia)

210. b — It aids in the removal of carbon dioxide (CO_2)

211. b — Patient at risk of aspiration

212. a — Injury from excessive tidal volume and airway pressure

213. a — Incomplete exhalation leading to positive pressure build-up

214. d — External Ventilation Mask

215. b — Respiratory failure due to airway constriction

216. b — To provide full ventilator support if the patient cannot breathe on their own

217. b — Synchronous Intermittent Mandatory Ventilation (SIMV)

218. c — A patient who can breathe independently but needs assistance with the effort

219. b — It keeps the airways open without controlling the patient's breaths

220. b — The ventilator provides all breaths with no spontaneous breathing allowed

221. d — Increased oxygen saturation

222. b — To prevent the development of stress ulcers

223. d — All of the above

224. a — High levels of oxygen delivery

225. c — A patient with severe airway obstruction or respiratory failure

226. c — Facilitates carbon dioxide removal

227. a — High PEEP and excessive tidal volume

228. c — Are starting to breathe independently but need some ventilator assistance

229. d — Acute respiratory distress syndrome (ARDS)

230. a — Assist/Control (A/C)

231. b — Oxygen toxicity

232. b — To assess whether sedation can be reduced or stopped

233. b — Pneumothorax

234. b — High Frequency alarm

235. c — Transfer the patient to advanced critical care

N1. b — Moderate ARDS
Berlin: mild 200–300, moderate 100–200, severe <100.

N2. b — Secretions, kinking or biting the tube
Low-pressure alarms indicate leaks/disconnection.

N3. a, b, c, d — HOB 30–45°; Daily sedation interruption; Oral care with chlorhexidine; Peptic ulcer & DVT prophylaxis
Circuits are changed only when soiled.

N4. b — 4–6 mL/kg
Low VT, plateau pressure <30, PEEP, FiO2 <60%.

N5. b — Intermittently during withdrawal, ≤10–15 s, at 80–120 mmHg
Pre-oxygenate and limit duration to prevent hypoxia.

N6. b — Thrombolytic therapy
Stable PE: anticoagulation; unstable: thrombolysis.